React.js

MY LEARNING NOTES

Handwritten-style visual edition

WHAT'S INSIDE

JavaScript foundations • Components • JSX • State • Props • Forms • Routing • API/Axios • useEffect

Content is preserved from the original notes; only the visual presentation has been redesigned.

1 mutable vs Immutable

```
<!-- array objects destructuring import and export  -->
<!-- map filter arrow fun(implicit function) spread opreator  -->
```

```
var arr = [1,2,3,4,5,6,'Ashu',7.1]
```

```
>React mai apko ek state naam ki chiz milegi jisko aap mutate directly nhi kar skte
```

or add nhi kar sakte

iske liey apko immutable way sai kam karna apadgea matlab aap naye jagah store karogey

```
let arr = [1,2,3,4,5]
```

arr.pop() // yeh react allow nhi karta directly changed

app yeh kar sakte ho

```
let arr=[1,2,3,4]
```

arr=[1,2,3]

2 spread operator use for copy thing

```
let arr = [1,2,3,4]
let copy = [...arr]
```

copy.pop()

output:- [1,2,3]

arr

output :- [1,2,3,4]

```
var obj = {name:"Ashutosh",age:"19"}
```

obj.name="dev" doesn't allow

```
var copy = {...obj}
```

copy.name="dev"

state=copy

state

Output:- {name:'dev',age:'19'}

```
>React mai hum directly chnage nhi karte copy bana kai change karte hai
```

3 Destructuring

Suppose there is a object and if we want to access there property n times it create difficulty for developer so we use shotest way which you see in example:-

eg-1

```
let obj ={name:"Ashutosh",age:19}
const {age} = obj
```

age -- we can easily access by writing property name --

output :- 19

eg-2

```
let obj = {
```

name : "Ashu",

Social:{

facebook:{

user:"alwaysashumishra",

gender:"Male"

```
    }
  }
}
```

```
const {user} = obj.social.facebook
```

user

4 yeh sab yeh batate hai ki hume itna lamba tarikaa nhi use karna hai access karne kai liye we can easliy access buy using some technique.iski ko kaihte hai destructuring

```
let arr = [12,function(){},13];
```

```
let [ first , , second]= arr
```

first

Output:-12

5 Import and Export

hum log component banate hai , component matlab page ka hissa navbar , sidebar , landing ,abh dikaat yeh kki har hissa alag alag component hai sab ki alag alag file hai last mai sabko jodna bhi padta hai,jodne kai liye use hota import

navbar - export

sidebar - export

cart - export

main- import navbar , sidebar , cart

-> component ek function hote hai unko hum create karne kai baad export karte hai fir dusri file mai import karet hai How?

eg-

```
function car(){
```

return

```
}
export default car;
```

--then in other file we import this--

```
import car from "./components/car"
```

eg - we can export multiple function though file

by using export keyword before function

```
export function cart(){
```

return

```
}
```

```
export function navbar(){
```

return

```
}
```

```
export default cart,navbar
```

--then in other file we import this --

```
import {cart,navbar} from "./components"
```

6 arrow function

```
const abcd = (val)=>{
```

console.log(val);

```
}
```

abcd(val);

if function have one parameter then you can remove bracket

```
 const abcd = val=>{
```

console.log(val);

```
}
```

abcd(val);

```
const abcd = (val1,val2)=>{
```

console.log(val);

```
}
```

abcd("Ashu","mishra");

output:-

Ashu mishra

7 arrow function with implicit return

```
const abcd = ()=>12;
```

console.log(abcd()+1);

OUTPUT:-

13

IF WE PUT BRACKET AFTER ARROW THEN VALUE DOESN'T RETURN

8 if we want to return directly object through function we put the object in these braket ()

```
const abcd = ()=>({
```

name:"Ashutosh",

age:20

```
})
```

9 Why we use map ? map hum log isliye use karte hai ki yeh naya aray create karke deta hai jaishe hume ek array kai har elment mai +1 karna hai lekin hum directly nhi kar sakte kyo ki react allow nhi karta hai toe hum map kai through har element mai +1 bhi kar sakte hai or naya array create hota hua milega

```
let state = [1,2,3,4,5]
```

```
let ans = state.map(val=>val+1);
```

state= ans;

10 jab original array ka size na kam karna ho waha map use hota hai

```
<!-- jab original array ka size agr kam karna ho toe waha filter use hota hai -->
```

```
let arr = [1,2,3,5,6];
```

```
let ans =arr.filter(elem=>elem>3);
```

React is a libraray which is inveneted by facebook engineer because in face book if notication ati thi toe vo page reload karne pai dikhthi thi toe ishi ko solve karne kai liye react.js banayigayi jeishme jaishe notifiction aye toe dikhe bina page reload kiye

or suppose there is website which is made of html css and js if you do change in particular text content of the website then after this happen is that every thing in website get repaint during updation of the text so this make website slow some time website get crash , so to solve this react.js is used in react there is vdom in which only that element is change which we want to change no other element get change

11 setup the react.js

1.tabnine extension is used because it give suggesion code .

2.prettier extension is used because it make format code .

3.node.js it is run time enviourment

4.vite vs crt

react mai kya hota that ki create react app hota tha jiske liye code likhna padta bahut sara abh nayi approach kai doran hum next with react.js karne lage thai jishe bahut log ko nhi pasand araha tha toe logo ko react raw hi sikhni thi toe ush samay vite naam ka tool wild cart entry lete hue kkaitha ki mere pass raw react hai aao chalo mere ko use karke kam karte hai

hum create-app kai through phele pure react banate thai lekin abh jabse ushme kuch chije add hogayi hai logo maja nhi araha toe vite tool ko use karke aap poorana react sikh pao gey

abh hum vite kai through app ko create karengey

12 the followimg step to create react app

1.Go in terminal .

2.then cd desktop

3.npm create vite@latest

4.the give app name

vanilla :- koi bh react nhi koi bhi framework nhi only html css

```
jsx => JavaScript XML is atually looking like a html but with having superpower
```

example:-

1. <h1>hey 2+2</h1>

output:- hey 2+2 because html not do any calculation

2. <h1>hey {2+2} </h1> convert to react.create element mai

output:- hey 4 output beacuse it is xml form in curly bases operation held

13 1. What is component ?

-- When you see side bar in facebook here is small component are there which looking like same but having different data in each particular component but you think that these component are individually made that is wrong because react make only one component with empty data filling that component use multiple time and just like post of facebook in this one component made which called post in this component data is change no need to made multiple div.

14 2. how to create component ?

1.function component :- When function return jsx that jsx convert to react.create element then is called function.

```
import React from "react";
```

```
function App(){
  return (
    <h1>hey</h1>
```

)

```
}
```

Export default app;

```
<!-- why we write import React From "react"--> because when function return the jsx code then  jsx code convert into React.create element then com
```

to open terminal shortcut key is (ctrl+`)

15 4. how code is running this to be understand

all components:-

1.header --> import inside app.jsx where website is goes to ready

2.landing --> import inside app.jsx where website is goes to ready

3.footer --> import inside app.jsx where website is goes to ready

then , after this app.jsx handle by main.jsx then main.js put app.jsx inside root div in html file.

16 4. Most use word in react is components, props,state,useEffect

--Components :- the components is used when you repeat some thing or you want to reused something suppose there is flipkart where product is show in particular block and you notice that block looking same but data is different so the reason behind this we make one dummy coomponent just like block but in this we not give real data to dummy components .

```
=> how components are made ?
```

- first we make components folder, inside in it make components like navbar, about etc.
- then these component are import in app.jsx
- we return component<navbar/> like this.
- return kai agey() braket isliye lagate ki koi sentence braket kai under likh sakte ho kisi bhi line mai

- return kai agey() braket ni lagaogey toe kisi bhi line ko return sai niche nhi --- likhogey nhi toe error show karega

- ek sath ek sai jaida return nhi kar sakte hai

- prewritten component lane kai liye write rafce or rfce

17 1. npm install tailwindcss @tailwindcss/vite

18 1. import tailwindcss from '@tailwindcss/vite'

2. plugin the tailwindcss() in same file

```
export default defineConfig({
```

plugins: [

tailwindcss(),

```
],
});
```

:- then import tailwind in css file

1. @import "tailwindcss";

19 1. npm run dev

20 5. making components

In this we make component

6. components reusability

In this we reuse components but not duplicate the components

hum log component mai return kai phele code likhengey toe vo run hoga phele , so creator kya karte hai , data ko return kai phele likhte hai fir return kai under use karte hai

21 When we see lots of post then we think that these post is made every time for every different data , then answer is No

Suppose , When these post are make using html, css, js only then it is true that for different data different post are made mean lots of div are made in line then we use react what is different the diffrence is that these post are made buy some other technique means we don'tmake lots of div for every post we make only one component of post in which we not give real time data we give temproray data, for every temporary data the component is return ,here not duplicacy is there only reuse one component again for diffrent data.

```
function App() {
```

```
const data = ["Amazon","Flipkart","Messho","Jiomart"]
```

```
return (
  <div>
      {data.map((elem, index)=>(
       <div key={index} className='w-40 h-40 w-fit bg-zinc-300'>{elem}</div>
```

))}

```
  </div>
```

);

```
}
```

```
export default App
```

22 7. Conditional Rendering :-

kis tarike sai content ko show karna chate ho

suppose button show this is instock and hamesha instock rahega nhi kabhi toe out tock rahega tab vo show karna padega

toe hum instock variable banyengey usko true ya false store karengey if true hoga toe button wali jagah pai instock dikhangey agr false toe outstock hi dikhangey mai chaita ho jab instock ho toe color button ka aglag show kare and jab outstock hoe to tab button color alag show kare agr kisi ek class ko change karna ho toe double quotes hata denegy or classes ko curly beses kai under back tick kai under rakhengey

double quotes tab hatana jab ek class ko alter karna ho fir sabhi class ko curley braces kai under back tick lagake unko waha rahkengey zishe hum string pas kar sake the ${} iske under vo class pai update karnegy zispe hume karna tha logic likhte hai

```
<button className={`px-4 py-1 ${elem.instock? 'bg-amber-200' : 'bg-blue-500'}  rounded text-black text-xs mt-2`}>{elem.instock ? 'instock' : 'out
```

```
import React from 'react'
```

```
function Card(){
```

//array of objects

```
const data = [
  {image:'https://images.unsplash.com/photo-1633174524827-db00a6b7bc74?q=80&w=896&auto=format&fit=crop&ixlib=rb-4.1.0&ixid=M3wxMjA3
  },
  {image:'https://images.unsplash.com/photo-1654573817889-296cad084c97?w=400&auto=format&fit=crop&q=60&ixlib=rb-4.1.0&ixid=M3wxMjA3
  {image:'https://i.pinimg.com/736x/7c/2e/d6/7c2ed668416881d16615000622f6b0ad.jpg',name:'Meesho',description:'Lorem ipsum dolor sit amet cor
  {image:'https://i.pinimg.com/736x/15/42/92/154292f7cd1b18711456e60b022a4095.jpg',name:'Jio Mart',description:'Lorem ipsum dolor sit amet cor
 ]
```

```
return (
  <div className='w-full h-screen flex items-center justify-center gap-20  bg-zinc-200'>
     {data.map((elem,index)=><div key={index} className='w-52 bg-zinc-100 rounded-md overflow-hidden' >
        <div className='w-full h-32 bg-zinc-300'>
           <img  className="w-full h-full object-cover" src={elem.image} alt="" />
        </div>
        <div className='w-full px-3 py-4'>
           <h2 className='font-semibold'>{elem.name}</h2>
           <p className='text-xs mt-3'>{elem.description}</p>
           <button className={`px-4 py-1 ${elem.instock? 'bg-amber-200' : 'bg-blue-500'}  rounded text-black text-xs mt-2`}>{elem.instock ? 'insto
        </div>
     </div>)}
  </div>
```

)

```
}
```

```
export default Card
```

23 8. Handling Events

In react if suppose there is button on click event is held that event not handle by queryselector or by another keyword that event is handle on button if button click then there is function which give respond after clicking button

```
 const songeventhandler = ()=>{alert('Downloading..')}
<button onClick={songeventhandler}className='px-2 py-1 bg-blue-500 text-zinc-100 text-xs rounded mt-2' >Download Now</button>
   </div>o
```

24 important is that if we put function inside onclick than we can't use function with braket if we use the automatic function get call, not call after click it is call before clicking on button

25 9. State management useState() hook

-BASIC :--

state is variable in which data is store and that data show in page if you want to change old data and then store and show new data in page the react

*we can put any data in useState().

*useState always a array deta hai.
Eg- const ans = useState(12);
ans[0] = value that you give in useState
ans[1] = update function that update value of useState

but in industry we write code by other way not like above code because the above code increase bulkness of code

Eg - const [a , b] = useState(0);
a - access the value which is in useState
b - you can update useState value by b and b is function.

26 Suppose we make a state variable whose name is score where value is store and that variable is used for showing data that variable have and that variable put in <h1> {Score} </h1> and also there is button which change the data of score by clicking on button <button onClick ={setscore}>change</button> how ? for this we use setcore function which update the value of score . then remind this we can't write function setScore() like this inside onclick beacuse it automatic change the value before clicking on button

27 you can't show value of boolean in h1 or inside any tag you have to convert that boolean to string using toString()

Eg-

```
const [banned,setbanned] = useState(true);
```

```
return (
  <div>
  <h1>{banned.toString()}</h1>
  <button onClick={()=>setbanned(!banned)}  className='px-2 py-1 bg-amber-300 text-amber-800 text-xs'>change</button>
 </div>
```

);

ye sab hum document.queryselector ko use karke event ko kyo handle nhi kar rahe hai

because react ishe bahut buri practice mnata hai vo bolata hai jo change karna hai useState mai change karo mai sabse main change kardunga

-Intermediate:--

dekho agr app chate ho ki button dubane per previous value ka ek increment wali value har bar mile then kya karengey hum ek variable banayengye

* code: -

28 const [val,setval]= useState(12);

return(

```
<h1>{val}</h1>
<button onClick= {()=>setval(prev=>prev+1)}  className='px-3 py-1 bg-blue-400 text-xs text-zinc-100 '>next</button>
</div>
```

)

-Avanced :--

```
* in we make varibale obj and store the object thorugh useState in obj then we make  button on clicking object property is changed
```

```
const [obj,setobj]= useState({name:"Ashutosh" , isbanned:true})
```

```
<h1>name :{obj.name}</h1>
 <h1>banned: {obj.isbanned.toString()}</h1>
 <button onClick={()=>setobj({...obj,isbanned : !obj.isbanned})}  className={`px-2 py-1 ${obj.isbanned ? 'bg-blue-200':'bg-red-500'}  text-ambe
 </div>
```

-Master level 1 :- add property in object which stored in state

* we make a state variable and stored a object in it with property name and age but if we want to add new property which is gender on clicking button then

```
const [object,setobject] = useState({name:"ashutosh",age:19})
```

```
<button onClick={()=>{
  setobject({...object,gender:'male'})
```

console.log(object)}}>click</button>

```
</div>
```

```
useState state turant ko update nhi karta hai , wo useState state apne hisaab sai
```

update karta hai , vo check karta hai ki newstate ,oldstate kahi barabar tioe nhi hai nhi toe fir vo rendring nhi karega

-Master level 2:- (onclick array last element get removed)

* In this we stored a array in state and then that array each element will be print and then we make button on clicking button the last element of particular array get delete for this we write code like this

```
const [arr, setarr] = useState([1, 2, 3, 4, 5]);
```

```
{arr.map(item=><h1>{item}</h1>)}
```

```
<button onClick={()=>setarr(()=>arr.filter((elem,index)=>index!=arr.length-1))}className="px-3 py-1 bg-red-600 text-xs text-white rounded-m
```

```
-Master level 3 :-(onClick array 2nd element removed)
```

* In this we stored a array in state and then that array each element will be print and then we make button on clicking in button the array 2 element delete

code-

```
const [arr, setarr] = useState([1, 2, 3, 4, 5]);
```

```
{arr.map(item=><h1>{item}</h1>)}
```

```
<button onClick={()=>setarr(()=>arr.filter((elem,index)=>index!=2))}className="px-3 py-1 m-1 bg-blue-500 text-xs text-white rounded">tap</b
```

```
-Master level 4 :-( onClicking delete array element which dived by 2 and show that element which not dividedby 2 )
```

* in this we again stored array in state variable and then we print all array element then and again we make button on clciking button the element of array removed which easily divided by 2

the code :-

```
const [arr, setarr] = useState([1, 2, 3, 4, 5]);
```

```
{arr.map(item=><h1>{item}</h1>)}
```

```
<button onClick={()=>setarr(()=>arr.filter((elem,index)=>elem%2!=0))}className="px-3 py-1 m-1 bg-blue-500 text-xs text-white rounded">tap<
```

```
-Master level 5 :-( onClicking add elemnt to the array )
```

* In this we stored array in state then we print all element of array then we make button on clicking button a new element add in array it is easy to add element in array while removing them from array

the code:- e

```
 const [arr, setarr] = useState([1, 2, 3, 4, 5]);
```

```
 {arr.map(item=><h1>{item}</h1>)}
```

```
<button onClick={()=>setarr([...arr,7 ])}className="px-3 py-1 m-1 bg-blue-500 text-xs text-white rounded">tap</button
```

revision:-

```
import React, { useState } from 'react'

function App() {

  const[val,setval]=useState(0)

  const[Value,setvalue]=useState(true)

  const[obj,setobj]=useState({name:'Ashutosh',age:19})

  const[arr,setarr] = useState([1,2,3,4,5,6])

  return (
    <div className='p-5'>
       <h1>Value:{val}</h1>
       <button  onClick={()=>setval(prev=>prev+1)}  className='mt-4 mb-4 px-3 py-1 bg-blue-500 text-xs text-white rounded'>click</button>

       <h2>State:{Value.toString()}</h2>
       <button onClick={()=>setvalue(!Value)} className='mt-4 px-3 py-1 bg-blue-500 text-xs text-white rounded'>tap</button>

       <h2>name:{obj.name}</h2>
       <h2>age:{obj.age}</h2>
       <button onClick={()=>setobj({...obj,age:20})} className='mt-4 px-3 py-1 bg-blue-500 text-xs text-white rounded'>tap me</button>

       {arr.map(elem=><h1>{elem}</h1>)}
        <button onClick={()=>setarr(()=>arr.filter((elem,index)=>index!=2))} className='mt-4 px-3 py-1 bg-blue-500 text-xs text-white rounded'>ta

       {arr.map(elem=><h1>{elem}</h1>)}
        <button onClick={()=>setarr([...arr,7])} className='mt-4 px-3 py-1 bg-blue-500 text-xs text-white rounded'>clicked</button>
    </div>
)

}

export default App
```

29 useState hook

Now we all understand what is state ? and why we use state ?

now we again make state variable which stored some value then we use setstate function for updation of value

```
 const[val,setval]=useState(false)
```

```
<div className='p-5 h-screen bg-blue-100 '>
      <h1 className='border-2 p-2'>{val===false?'bahar jao':'mat jao'}</h1>
      <button onClick= {()=>setval(()=>!val)} className='mt-2 px-4 py-2 rounded-md text-xs bg-cyan-700 text-white '>update</button>
</div>
```

30 10. Passing data through props:-

suppose aapne ek component banaya hai or uske under ek button ziska kuch color hai or kuch text likha hai lekin zab tum usko main app.js file mai component ko use kar rahe diffrent jagah then tum sochte har jagah button same nhi rahena chahiye then tumne app.js pai jaha jaha component ko use kiya hai waha data dogey ki mujhe ish type ka text cchaiye or style of button ish type ki then yeh data zayega componet mai props parameter mai likhe zayengey zo hamare data ko baddlne madat karega har jagah

31 child component code:-

```
import React from "react";
```

```
function Card3({text,color}) {
  return (
    <div>
      <button className={`px-3 py-1 rounded-md text-xm ${color} text-zinc-100  m-2`}>{text}</button>
    </div>
```

);

```
}
```

```
export default Card3;
```

32 parent component code:

```
import React from 'react'
import Card3 from './components/Card3'
```

```
function App() {
  return (
    <div className='flex'>
```

```
     <Card3 text="know more" color="bg-blue-400"/>
    <Card3 text="click me" color="bg-red-400"/>
```

```
    </div>
```

)

```
}
```

```
export default App
```

* prop drilling usasge:-

humara data actually mai main app component mai hoga usko props ki madat sai pass krengy card main ,har card pai ek add friend button hoga and hume add friend button pai click karne pai alert dena hoga

33 11. Props drilling Exercise :-

34 In this we make app in which at top the navbar is there and at center musics are shown in which button are there inside button text is add to favorite on clicking button the music is added to favorite section and favorite value increase

35 12. form handling:-

generally ye hota ki jab hum form submit karte hai toe page reload hota hai ye nhi hona chahiye isilye toe react use karte vo reload ko allow nhi karta

36 1. useref.

2. controlled components.

3. react hook form (mostly used for hanlding submit reloading).

37 useref :-

ishme hum sabhi input ko select karlete hai or unki value submit hone kai baad nikhalte hai

* useref sai kisi bhi input ko select kar sakte ho

```
import React, { useRef } from 'react'
```

```
function App() {
  const name=useRef(null)
  const age=useRef(null)
  const email=useRef(null)
```

```
  const eventhandler=(event)=>{
```

event.preventDefault();

console.log(name.current.value,age.current.value,email.current.value

)

```
  }
```

```
  return (
    <form action="" onSubmit={eventhandler} >
      <input ref={name} type="text" placeholder='name'/>
      <input ref={age} type="text" placeholder='age' />
      <input ref={email} type="email"placeholder='email' />
      <input  type="submit" />
```

```
    </form>
```

)

```
}
```

```
export default App
```

38 controlled component:

app jabhi kuch likho real time pai usestate pai updater kar dein

jaishe kuch liko setstate kai through new wali value state mai store kar de

```
import React, { useState } from 'react'
```

```
function App() {
```

```
  const[val ,setval]=useState({name:"",email:""})
```

```
  const eventhandler=(event)=>{
```

event.preventDefault();

console.log(val)

```
  }
  return (
    <form action=""onSubmit={eventhandler} >
      <input type="text" onChange={(event)=>setval({...val,name:event.target.value})}placeholder='Enter your name' />
      <input type="email" onChange={(event)=>setval({...val,email:event.target.value})}
```

placeholder='Enter your email'/>

```
      <input type="submit"  />
    </form>
```

)

```
}
```

```
export default App
```

39 react hook form :-

In this we install react-hook-form and then import and then there is a register keyword of useform which use form registeration and other is handlesubmit which handle form after submit and write register in input section by which like this{...register('')} in this all function which carry by regiter are spread nd used

```
import React from 'react'
import { useForm } from 'react-hook-form'
```

```
function App() {
  const {register,handleSubmit} =useForm()
```

console.log(register)

```
  return (
```

```
    <form action="" onSubmit={handleSubmit((data)=>{console.log(data)})} >
      <input type="text" {...register("name")} placeholder='enter your name'/>
      <input type="email" {...register("email")} placeholder='...@gmail.com'/>
      <input type="submit" />
    </form>
```

)

```
}
```

```
export default App
```

40 React Routing

mai ye hota hai ki jaishe hume ek page sai durse page pai jana hai or dusre componets pai jana hai toe waha routing ki madat lagti hai

agr hume apne system mai routing apply karni hai toe hume react-router-dom ko use padega

terminal :-

1.npm install react-router-dom

2.hum jayengy main.jsx file mai ushme browserouter tag ko use karengey jo ki react router dom sai mila hai jishe hum pure application mai routing laga sakte hai

react mai aap anchor tag ka use nhi kar sakte ek page sai dusre page mai jane kai liye kyo ki anchor tag kya karta hai vo yeh karta hai jab hum ek page sai dusre mai jate hai then reloading start hoti hai jo ki bahut kharab hai toe react kya karta hai isiliye vo react router dom ko use karta hai or fir ushme link name ka tag hota hai jaishe ki anchor tag ushme reloading nhi hoti jab hum ek page sai dusre page mai jate hai yeh link tag react router kai under hota hai

toe hum link tag ka use karte hai jo ki react router dom mai paya jata haai hai usko phele import karengey then link use karengey uske badth ushme to =" "

kai under path fill karenegy jishe hum ush path per ja sake clcik karne pai then uske baad

* import React from 'react'

```
import { Link } from 'react-router-dom'
```

```
function Nav() {
  return (
    <nav className=' p-8 flex gap-20 justify-center bg-blue-100 text-xl'>
      <Link to="/home">Home</Link>
      <Link to="/user">User</Link>
      <Link to="/about">About</Link>
    </nav>
```

)

```
}
```

```
export default Nav
```

41 hum routes tag ko use karenegey uske phele hum usko import karengey then hum routes tag kai under route tag likhengey jishme paath fill kiya jata hai or element ko retun kiya jata hia jaaishe hi path match karta hai link wale path sai then hum return karte element or unh component ko import karengey jinko return kare rahe hai

```
import React from 'react'
import { Route, Routes } from 'react-router-dom'
import Home from '../components/home'
import About from '../components/About'
import User from '../components/User'
```

```
function Routing() {
  return (
    <Routes>
        <Route  path="/home"  element={<Home/>}   />
        <Route  path='/user'  element={<User/>}   />
        <Route  path='/about' element={<About/>}  />
    </Routes>
)
}
```

```
export default Routing
```

* jitna app.jsx pai kam code rakho gay utna accha hai

agr hum chate hai jaishe hi home pai click karey toe home wala page khule or wo alag color mai dikhai dei navigation kare toe uske liye hum link ki jagah navlink use karengey jo ki react router provide karta hai jishe hum style karke color laga sakte hai jab click ho color dikhayi dai

ishme ye hota jasihe jo link pai click hoga vo active hoti hai jab active tab color laga do

-- Navlink mai teen tarike sai highlight kar sakte hai

42 1. style

2. children

3. classes kai through

43 style

```
import React from 'react'
import { Link , NavLink} from 'react-router-dom'
```

```
function Nav() {
  return (
    <nav className=' p-8 flex gap-20 justify-center bg-blue-100 text-xl'>
      <NavLink style={(e)=>{
```

return{

color: e.isActive? 'tomato' : '',

fontWeight:e.isActive?'bold':'',

```
        }
      }} to="/home" >Home</NavLink>
      <NavLink style={(e)=>{
```

return{

color: e.isActive? 'tomato' : '',

fontWeight:e.isActive?'bold':'',

```
        }
      }} to="/user">User</NavLink>
      <NavLink style={(e)=>{
```

return{

color: e.isActive? 'tomato' : '',

fontWeight:e.isActive?'bold':'',

```
        }
      }} to="/about">About</NavLink>
    </nav>
```

)

```
}
```

```
export default Nav
```

44 classname ko use karke style dena navigation ko

```
import React from 'react'
import { Link , NavLink} from 'react-router-dom'
```

```
function Nav() {
  return (
    <nav className=' p-8 flex gap-20 justify-center bg-blue-100 text-xl'>
      <NavLink className={(e)=>{
```

return[

e.isActive? 'text-red-500' : '',

e.isActive?'font-bold':''

```
        ].join(" ")
      }} to="/home" >Home</NavLink>
      <NavLink className={(e)=>{
```

return[

e.isActive? 'text-red-500' : '',

e.isActive?'font-bold':''

```
        ].join(" ")
      }} to="/user">User</NavLink>
      <NavLink className={(e)=>{
```

return[

e.isActive? 'text-red-500' : '',

e.isActive?'font-bold':''

```
        ].join(" ")
      }} to="/about">About</NavLink>
    </nav>
```

)

```
}
```

```
export default Nav
```

45 How to integrate api in react.js?

```
* usage with api ajax (fetch/ axios)
```

how to integarte backend api in react ?

dekho kya hota hai react isliye used karte hai kyo ki yeh refersh nhi hone deta hai and data jo backend kai through ata hai frontend mai usko hum api kai through late backend data leta hai database sai and backend then integrate hota hai forntend kai sath then hum data ko showcase karte hai

* * jab koi bole ki backend create karna hai matlab backend or database kai bichme connection hona ziske through data ko bhejte hai databse mai or store karte hai or data ko database lete bhi hai and backend ko controler bhi bolte hai

dekho react mal hote hai bahut sare pages and ush page mai humko dta bhi dikhana hota hai uske liye hume database chahiye orlekin dtatabse tak pahuchne kai liye humko backend sai communication karni padti hai toe yeh jo

* { Connection hum banate hai backend and frontend kai biche mai uske liye ajox concept lagata hai jo ki databse sai data lata hai or frontend mai sh

Now we do connection between frontend and backend using concept of ajox and which is implemented by axios

backend jis chiz ki karan hum connect kar parahe hai usko hum bolte hai api (Application programming interface)

46 WHAT IS API?

An api is a link which provided by backend and when this api we call using axios then we go in backend and acces data as per requirement

We can get data from axios or we can send data through axios

200 mean complete status

47 so lets see in code how axios do wrok after applying ajax for connection then the link or api through which backend is connected to frontend that api is call by axios to access data from backend and show in frontend and axios is also use api to post some data to from frontend to backend

* axios access data from api calling using get() method
* axios send data through api to database using post() method in post method you have to inter your data in form object that you want to send

48 import React, { useState } from "react";

```
# import axios from "axios";
function App() {
  const [product, setproduct] = useState([]);
  const getproducts = () => {
    const api = "https://fakestoreapi.com/products";

      axios.get(api).then((products) => {
```

console.log(products);

setproduct(products.data);

```
      }).catch((err) => console.log(err));
  };

  const addproduct = () => {
    const api = "https://fakestoreapi.com/products";

      axios.post(api, {
```

id: 0,

title: "string",

price: 0.1,

description: "string",

category: "string",

image: "http://example.com",

```
      }).then((resp) => {
```

console.log(resp);

```
      }).catch((err) => console.log(err));
  };
```

console.log(product);

```
  return (
    <div className="p-3">
      <button onClick={getproducts} className="px-3 py-1 bg-red-200 ">
```

Call product Api

```
      </button>
      <br />
      <br />
      <button onClick={addproduct} className="px-3 py-1 bg-red-200 ">
```

post

```
      </button>
```

```
      <hr className="m-1"/>
      <ul className="p-1">
        {product.length > 0 ? (
          product.map((p,index) => (
            <li key={index} className="p-5 m-2 bg-red-300 rounded-md">{p.title}</li>
```

))

) : (

```
            <h1>loading...</h1>
```

)}

```
      </ul>
    </div>
```

);

```
  }
```

```
  export default App;
```

in above we calling api and get data that data has set of objects and we store that data in state varible which is array then we show that data in ui by applying map function in state varible duw to this each object show its own data independtly

49 Hook-useEffect() lifecycle method

if we want when we open application then are product will be there .

not after clicking on button .

```
  ye tab hoga jab jo getproduct wala function apne man sai chalega ye jo achieve hota hai useEffect ko use karke hota hai
```

50 1. Component mount - creation

2. Component Update

3. Component Unmount -deletion

that all for react component

Now Understand in breif way

```
  Fist we make component folder then in this we make show file in which we put the code which calling  api and get data and that data is store in useSta
```

we make it beacause we want to understand what is lifecycle method

when we open component like home then that component will create this is called component mount and if we switch to another component like services than services create home will delete that is called component deletion and if we open show and then service componet will delete show will create and if we do changement in show component like calling api usestate get changed is called component updation

`

```
* useState hook hume kisi component mai create karne ka or store karne ka facility deta hai
* useEffect hook hume kisi bhi components ki life cycle use karne ka awser deta hai
```

```
useEffect humari lifecycle method ka implementaion hai jo ki humare react kai duara diya jata hai or kab kab coponent create honepar delete  hone pa
```